PRACTICAL 2

Basic System Commands and Input/Output Redirection

Aim

To study basic Linux commands and demonstrate standard input, standard output, error redirection and pipes.

Commands

Display current directory

pwd

List files

ls

ls -l

ls -la

Display date

date

Display current user

whoami

Display logged-in users

who

System information

uname -a

Hostname

hostname

Input/Output Redirection

1. >

Creates/overwrites a file.

echo "Linux Practical" > practical.txt

cat practical.txt

Output:

Linux Practical

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2. >>

Appends data.

echo "TYIT RHEL Practical" >> practical.txt

cat practical.txt

Output:

Linux Practical

TYIT RHEL Practical

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3. Pipe |

ls -l | less

Another example:

ps aux | grep ssh

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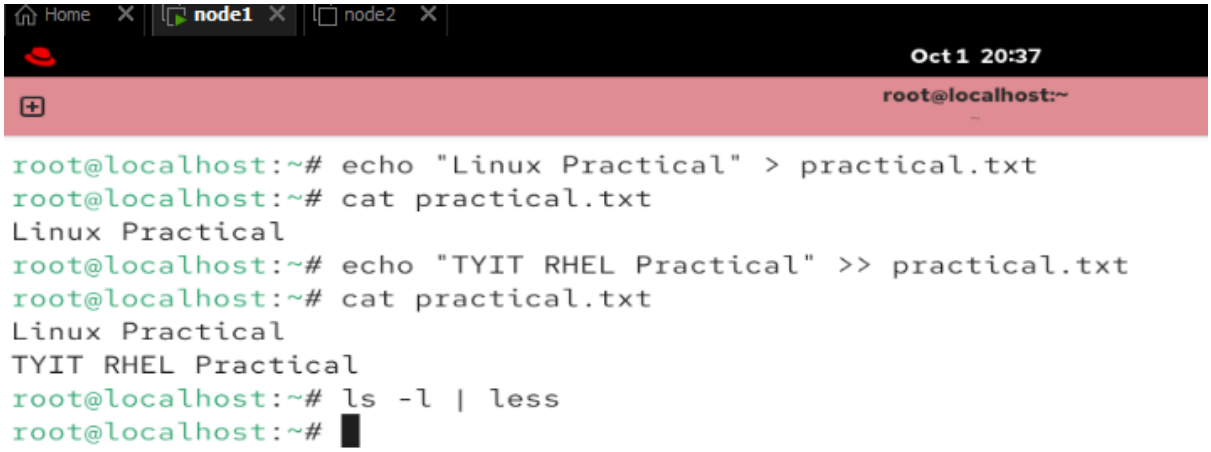
4. Error Redirection 2>

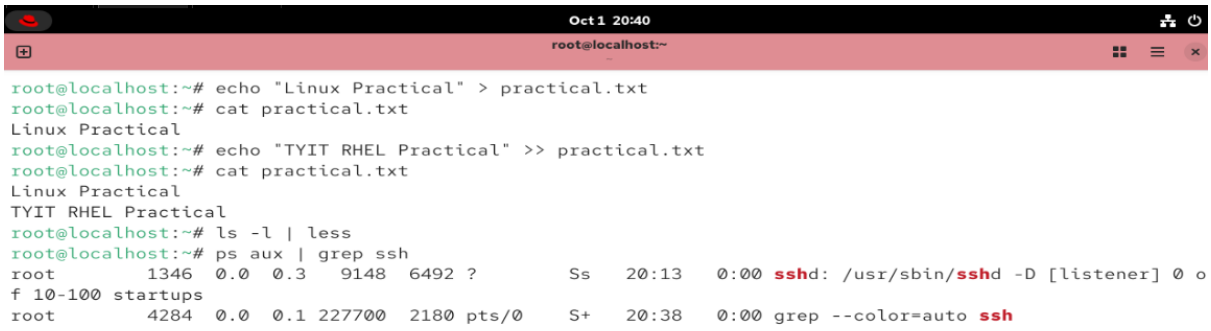
ls /wrong-directory 2> error.txt
cat error.txt

---

5. Redirect output and errors
command > output.txt 2> error.txt

Students should execute:
echo "Linux" > a.txt
echo "RHEL" >> a.txt
cat a.txt
ls /wrong 2> error.txt
cat error.txt





PRACTICAL 3

grep, Regular Expressions and Text Processing

Aim

To analyse text using grep, regular expressions and text-processing commands.

Create a file:

cat > students.txt

Enter:

Amit TYIT 85
Neha TYIT 92
Rahul SYIT 76
Priya TYIT 88
Rohan SYIT 69
Sneha TYIT 95

Press **Ctrl+D**.

---

grep

grep TYIT students.txt

Expected:

Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95

Case-insensitive grep

grep -i tyit students.txt

Line numbers

grep -n TYIT students.txt

Regular Expression

Find lines beginning with A:

grep '^A' students.txt

Find lines ending with 5:

grep '5$' students.txt

Find exactly three digits:

grep -E '[0-9]{3}' students.txt

Text File Commands

cat

cat students.txt

head

Your syllabus says hesd; the correct command is:

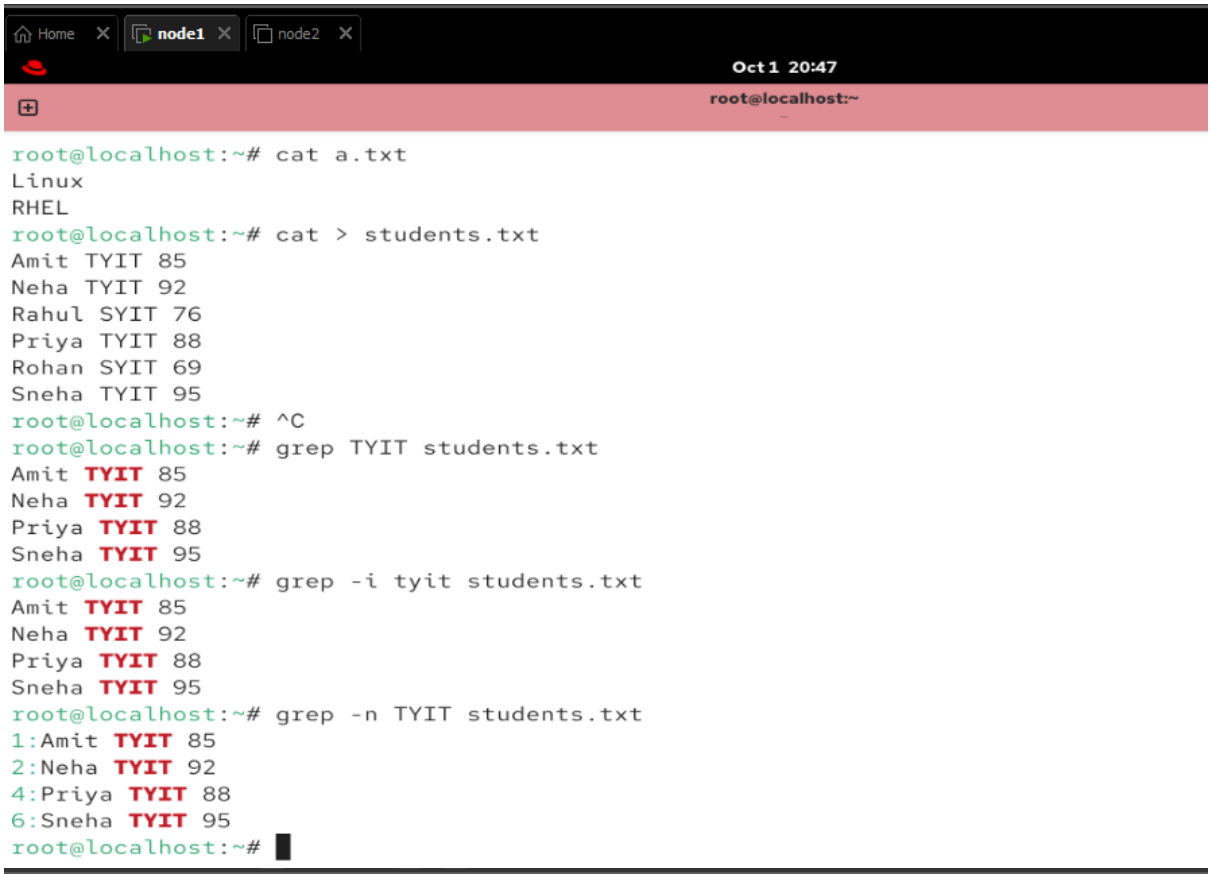
head students.txt

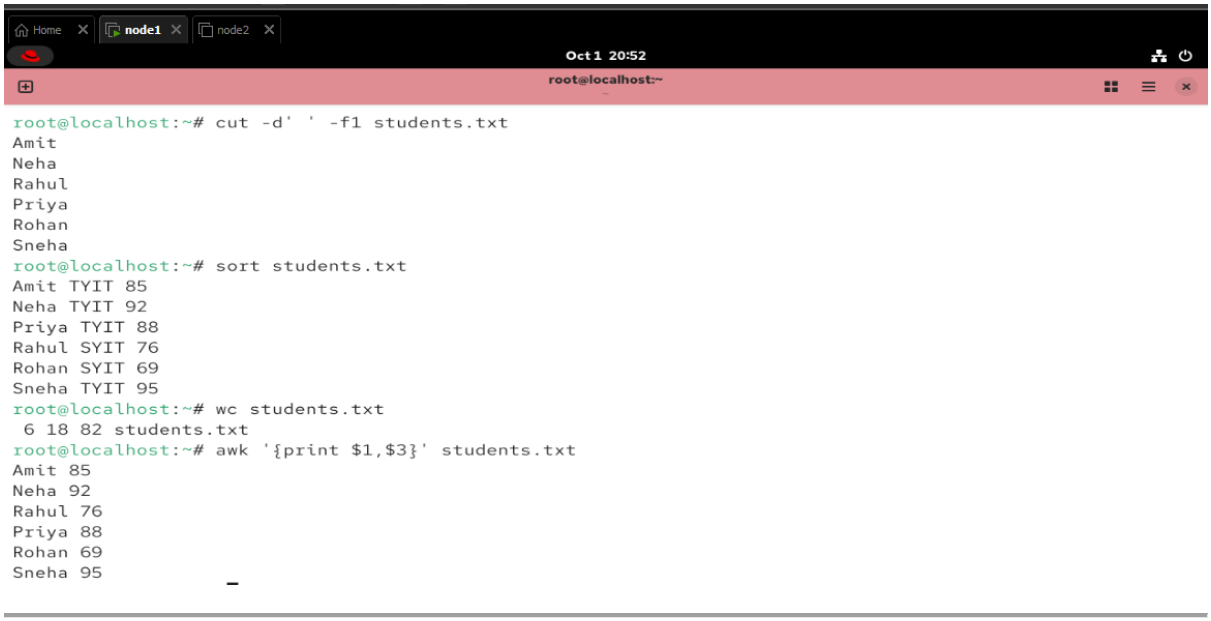
tail

tail students.txt

less

less students.txt
Press q to exit.
cut
cut -d' ' -f1 students.txt
sort
sort students.txt
wc
wc students.txt
awk
awk '{print $1,$3}' students.txt
Expected:
Amit 85
Neha 92
Rahul 76
Priya 88
Rohan 69
Sneha 95


Home
node1
node2
Oct 1 20:47
root@localhost:~
root@localhost:~# cat a.txt
Linux
RHEL
root@localhost:~# cat > students.txt
Amit TYIT 85
Neha TYIT 92
Rahul SYIT 76
Priya TYIT 88
Rohan SYIT 69
Sneha TYIT 95
root@localhost:~# ^C
root@localhost:~# grep TYIT students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95
root@localhost:~# grep -i tyit students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Sneha TYIT 95
root@localhost:~# grep -n TYIT students.txt
1:Amit TYIT 85
2:Neha TYIT 92
4:Priya TYIT 88
6:Sneha TYIT 95
root@localhost:~#


Home
node1
node2
Oct 1 20:52
root@localhost:~
root@localhost:~# cut -d' ' -f1 students.txt
Amit
Neha
Rahul
Priya
Rohan
Sneha
root@localhost:~# sort students.txt
Amit TYIT 85
Neha TYIT 92
Priya TYIT 88
Rahul SYIT 76
Rohan SYIT 69
Sneha TYIT 95
root@localhost:~# wc students.txt
6 18 82 students.txt
root@localhost:~# awk '{print $1,$3}' students.txt
Amit 85
Neha 92
Rahul 76
Priya 88
Rohan 69
Sneha 95